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Behavior Analysis of Fine Temporary Plugging Agents and Fibers in Bridging Regions: Migration and Clogging

Jinzhou Zhao, Yinjian Huang, Lan Ren, Ran Lin, and Zhihao Yu, State Key Laboratory of Oil and Gas Reservoir Geology and Exploitation, Southwest Petroleum University, Chengdu, China; Jianfa Wu, Yi Song, and Cheng Shen, PetroChina Southwest Oil & Gasfield Company, Chengdu, China

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Abstract

Temporary plugging and diverting fracturing enhances unconventional reservoir stimulation by creating a tight sealing layer through the bridging and filling of diverters in fractures. The tightness and pressure-bearing capacity of the sealing layer are governed by the clogging efficiency of fine particles and fibers in the bridging region. However, the clogging mechanism within the pores of coarse particles after bridging remains inadequately understood. This study develops a hybrid CFD-DEM method integrating resolved, semiresolved, and unresolved models. The method implements variable coupling precision for particles of different sizes within a unified fluid mesh, balancing the accuracy of the flow field solution and computational efficiency. Based on this method, the effects of particle size, concentration, and shape on the macroscopic transport and clogging behavior of diverters in the bridging skeleton are investigated. Furthermore, the sealing layer formation process is analyzed across multiple scales, from mesoscale clogging evolution to microscale interparticle contact characteristics. The results demonstrate that fine particles initially migrate rapidly along preferential flow paths within the bridging skeleton. As injection proceeds, smaller particles exhibit greater mobility and are less prone to clogging at pore throats, whereas larger particles tend to accumulate and form size-dominated clogs, significantly hindering subsequent particle transport. In mixed particle systems, small particles help alleviate clogging at pore throats and promote a more uniform spatial distribution, while large particles facilitate the formation of dense local clogging structures. Increasing particle concentration intensifies competition at pore throats and accelerates the development of dense and stable clogging in the lower part of the skeleton. Although fibers are composed of small constituent particles, their high aspect ratio restricts migration, leading to retention near the skeleton entrance and the formation of extensive U-shaped clogging regions. At low concentration, the normal contact force distribution in fine particle clogging zones is highly anisotropic. In contrast, higher concentrations, mixed particle systems, and the presence of fibers all reduce this anisotropy, thereby enhancing the structural stability and sealing performance of the clogged regions. This study reveals multiscale mechanisms of tight sealing formation in fractures and establishes a theoretical foundation for optimizing temporary plugging and diverting fracturing techniques. Moreover, the developed hybrid CFD-

DEM approach holds broader applicability in petroleum engineering scenarios involving fine powders or cross-scale particle transport.

Keywords: temporary plugging, fine particles, fibers, hybrid CFD-DEM, clogging

Introduction

With the depletion of conventional oil and gas resources, unconventional resources have become an important alternative energy source (Jia et al., 2023). China possesses substantial shale and tight oil and gas reserves with significant development potential. Multi-cluster staged fracturing in horizontal wells is essential for the efficient development of these unconventional reservoirs (Zhao et al., 2022). However, low reservoir quality and strong heterogeneity often lead to rapid production decline and low recovery rates (Yang et al., 2024). Temporary plugging and diverting fracturing techniques, which are implemented by injecting fracturing fluids containing particles, powders, or fibers (Chang et al., 2024), can create sealing layers in targeted zones. This improves reservoir stimulation uniformity, enhances perforation cluster efficiency, and increases fracture complexity. The success of this technique depends on whether diverter materials can be transported and can form a dense sealing layer at the target location. Therefore, a comprehensive understanding of the migration and clogging mechanisms of diverters in fractures is critical for optimizing unconventional reservoir stimulation.

Diverters are typically manufactured from degradable polymeric materials, which provide high mechanical strength, good degradation performance, and adequate formation adaptability (Zhu et al., 2021). They can be engineered in various sizes and shapes to suit different applications, with particles and fibers being the most widely used types. In fractures, diverters initially form a stable bridging skeleton through particle or fiber bridging, which serves as the basis for subsequent sealing layer formation (Chen et al., 2024). Many experimental and theoretical studies have shown that particle size, concentration, and shape directly influence bridging probability and pressure-bearing capacity: higher particle size to fracture width ratios and increased concentrations both promote bridging (Sun et al., 2019; Wang and Huang, 2023). The addition of fibers can further reduce the critical concentration needed for particle bridging (Potapenko et al., 2009) and improve the stability of the bridging structure. However, bridging alone does not fully block fluid channels, as ongoing seepage through skeleton pores often limits the tightness and effectiveness of the temporary plugging layer (Cohen et al., 2010). Further injection of fine particles or fibers into the bridging skeleton can fill residual flow channels, reducing the porosity and permeability of the sealing layer and improving its compactness and plugging performance (Cortez-Montalvo et al., 2015; Zhang et al., 2019). In these processes, coarse particles preferentially bridge in fractures, thereby providing surfaces that facilitate the accumulation and clogging of fine particles or fibers, which increases overall flow resistance. Although fine particles and fibers are important in forming dense sealing layers, their migration, accumulation, and clogging mechanisms at the pore scale are not fully understood. High- resolution numerical simulations are therefore necessary to clarify these processes and to support further optimization of temporary plugging and diverting fracturing.

The formation of a dense sealing layer in fractures involves hydraulic transport, collision, bridging, accumulation, and clogging of diverters. To study the formation mechanisms at mesoscopic and microscopic scales, fluid-particle coupled numerical simulation methods are essential for understanding the behavior of fine particles and fibers in the bridging region. Mainstream approaches include the coupled computational fluid dynamics and discrete element method (CFD-DEM) and the lattice Boltzmann method and discrete element method (LBM-DEM) (Li et al., 2023; Xiong et al., 2023). As a Lagrangian method, DEM effectively captures particle migration, accumulation, and clogging (Cundall and Strack, 1979). LBM-DEM is effective in handling complex boundaries but requires high computational costs for large-scale particle simulations (Doan et al., 2025). CFD-DEM treats the fluid as a continuous phase and particles as a discrete phase, calculating fluid-solid interactions through momentum exchange. Depending on the solution accuracy and grid-to- particle size ratio, models are classified as resolved, semi-resolved, or unresolved, with accuracy and computational cost decreasing accordingly (Cheng et al., 2018; Xie et al., 2023). Due to its balance of efficiency and accuracy, unresolved CFD-DEM is widely used to simulate proppant transport, bridging and plugging, sand production, and related processes (Kazidenov et al., 2023; Wang and Huang, 2023; Zhu et al., 2023). However, in the bridging region, clogging is governed by the pore-scale seepage field of the skeleton, and conventional CFD-DEM methods may not accurately capture the seepage-induced migration and clogging of fine particles and fibers with sufficient accuracy. Therefore, high-resolution pore-scale flow field simulations are required to study these mechanisms.

In this work, we propose a hybrid CFD-DEM approach that integrates resolved, semi-resolved, and unresolved models, enabling different coupling accuracies for particles of various sizes within the same fluid grid. This approach balances flow field accuracy with computational efficiency. We use this method to investigate the effects of particle size, concentration, and shape on the macroscopic migration and clogging behavior of diverters within the bridging skeleton, thus providing new insights into the formation of dense sealing layers in fractures.

Hybrid CFD-DEM method

Fluid phase

The hybrid CFD-DEM model employs resolved, semi-resolved, and unresolved approaches to solve the coupling between fluid and particles of different sizes. In the resolved model, the Fictitious Domain Method (FDM) is used to analytically solve the seepage flow field within the pores of the coarse particle skeleton (Shirgaonkar et al., 2009). In the fictitious fluid domain, the velocity of each fluid cell is constrained to match the motion of the corresponding coarse particle, thereby implicitly enforcing a no-slip boundary condition at the fluid-particle interface (Wang et al., 2022). Fluid flow in the entire region satisfies the continuity and momentum equations as follows:

$$\nabla \cdot \boldsymbol{u}_\mathrm{f} = 0, \ \text{ in } \Omega \tag{1}$$

$$\rho_\mathrm{f}\left(\frac{\partial \boldsymbol{u}_\mathrm{f}}{\partial t} + \boldsymbol{u}_\mathrm{f} \cdot \nabla \boldsymbol{u}_\mathrm{f}\right) = -\nabla p + \nabla \cdot \boldsymbol{\tau} + \rho_\mathrm{f}\boldsymbol{g} + \boldsymbol{F}_\mathrm{fd}, \ \text{in } \Omega \tag{2}$$

$$\boldsymbol{u}_\mathrm{f} = \boldsymbol{u}_\mathrm{p} + \boldsymbol{\omega}_\mathrm{p} \times \boldsymbol{r}, \ \text{ in } \Omega_\mathrm{cp} \tag{3}$$

Where $\boldsymbol{F}_{\mathrm{fd}}$ is the virtual body force introduced by mapping the particle velocity; Ω is the computational domain; $\boldsymbol{U}_\mathrm{p}$ is the particle velocity; $\omega_\mathrm{p}$is the particle angular velocity; $\boldsymbol{r}$ is the position vector from the particle center to the fluid cell; Ω_{cp} is the domain occupied by coarse particles.

The force exerted by the fluid on particles in the resolved model is calculated by volume integration of the fluid stress within the particle region based on Gauss's theorem (Xie et al., 2023), as shown below:

$$\boldsymbol{F}_\text{f-p} = \iiint_{\Omega_\mathrm{cp}} (-\nabla p + \nabla \cdot \boldsymbol{\tau}) dV \tag{4}$$

$$\boldsymbol{M}_\text{f-p} = \iiint_{\Omega_\mathrm{q}} \boldsymbol{r} \times (-\nabla p + \nabla \cdot \boldsymbol{\tau}) dV \tag{5}$$

Where Mf-p is the torque exerted by the fluid on the coarse particle.

The unresolved model utilizes local averaging theory to solve the Navier-Stokes equations and obtain the averaged flow field. Fluid-particle interaction forces are calculated using empirical models, enabling local-scale fluid-solid coupling (Zhou et al., 2010). The governing equations for the fluid phase are:

$$\frac{\partial(\varepsilon_\mathrm{f})}{\partial t} + \nabla \cdot (\varepsilon_\mathrm{f} \boldsymbol{u}_\mathrm{f}) = 0 \tag{6}$$

$$\frac{\partial(\varepsilon_\mathrm{f}\rho_\mathrm{f}\boldsymbol{u}_\mathrm{f})}{\partial t} + \nabla \cdot (\varepsilon_\mathrm{f}\rho_\mathrm{f}\boldsymbol{u}_\mathrm{f}\boldsymbol{u}_\mathrm{f}) = -\varepsilon_\mathrm{f}\nabla p + \varepsilon_\mathrm{f}\nabla \cdot \boldsymbol{\tau} + \varepsilon_\mathrm{f}\rho_\mathrm{f}\boldsymbol{g} - \boldsymbol{F}_\text{f-p} \tag{7}$$

Where ε_f is the porosity; $\boldsymbol{U}_f$ is the fluid velocity; p_f is the fluid density; t is time; p is pressure; $\boldsymbol{\tau}$ is the stress tensor; $\boldsymbol{g}$ is gravity; $\boldsymbol{F}_{\text{f-P}}$ is the total force exerted by the fluid on the particles.

The semi-resolved model uses the same governing equations as the unresolved model but differs in the computation of local porosity and the fluid field within the particle influence region. Specifically, the particle volume fraction at a given location is computed by volume-weighted summation of each particle at that point, which is then used to calculate the grid cell porosity. A Gaussian function is adopted as the weighting kernel to calculate the local fluid velocity, relative velocity, and porosity in the particle influence region (Wang et al., 2019).

$$\boldsymbol{u}_{\mathrm{f}}=\frac{\sum_{j=1}^{n}\boldsymbol{u}\left(\boldsymbol{r}_j\right)W\left(\boldsymbol{r}-\boldsymbol{r}_j, d_{\mathrm{p}}\right)\Delta V_{\mathrm{cell},j}}{\sum_{j=1}^{n}W\left(\boldsymbol{r}-\boldsymbol{r}_j, d_{\mathrm{p}}\right)\Delta V_{\mathrm{cell},j}} \tag{8}$$

$$\boldsymbol{u}_r=\boldsymbol{u}_{\mathrm{p}}-\frac{\sum_{j=1}^{n}\boldsymbol{u}\left(\boldsymbol{r}_j\right)\exp\left(-\frac{\left|\boldsymbol{r}-\boldsymbol{r}_j\right|^2}{2(\kappa d)^2}\right)\Delta V_{\mathrm{cell},j}}{\sum_{j=1}^{n}\exp\left(-\frac{\left|\boldsymbol{r}-\boldsymbol{r}_j\right|^2}{2(\kappa d)^2}\right)\Delta V_{\mathrm{cell},j}} \tag{9}$$

$$\tilde{\varepsilon}_{\mathrm{f}}=\frac{\sum_{j}^{n}\Delta V_{\mathrm{cell},j}\varepsilon_{\mathrm{cell},j}}{\sum_{j}^{n}\Delta V_{\mathrm{cell},j}} \tag{10}$$

Where n is the number of fluid cells in the particle influence region; $\boldsymbol{r}_j$ is the position vector of point j; W is the Gaussian kernel; $\Delta V_{\mathrm{cell}j}$ is the volume of cell j; k is the scalar coefficient of the kernel; $\tilde{\varepsilon}_{\mathrm{f}}$ is the porosity of the influence region; $\varepsilon_{\mathrm{cell,j}}$ is the porosity of cell j.

When computing the total force exerted by the fluid on particles in the semi-resolved and unresolved models, the main interaction forces considered are drag, pressure gradient, viscous force, virtual mass force, and Saffman lift, while minor contributions such as the Basset force are neglected. Drag force, the primary component of fluid-particle interaction in solid-liquid systems, is calculated using the Gidaspow model (Gidaspow, 1994):

$$\boldsymbol{F}_{\mathrm{d}}=\frac{\beta_{\mathrm{d}}V_{\mathrm{p}}}{\left(1-\tilde{\varepsilon}_{\mathrm{f}}\right)}\left(\boldsymbol{u}_{\mathrm{f}}-\boldsymbol{u}_{\mathrm{p}}\right) \tag{11}$$

$$\beta_{\mathrm{d}}=\begin{cases}\frac{3\rho_{\mathrm{f}}\tilde{\varepsilon}_{\mathrm{f}}\left(1-\tilde{\varepsilon}_{\mathrm{f}}\right)}{4d_{\mathrm{p}}}C_{\mathrm{d}}\left|\boldsymbol{u}_r\right|\tilde{\varepsilon}_{\mathrm{f}}^{-2.65}, \tilde{\varepsilon}_{\mathrm{f}}\geq 0.8\\ \frac{150\left(1-\tilde{\varepsilon}_{\mathrm{f}}\right)^2\mu}{\tilde{\varepsilon}_{\mathrm{f}}d_{\mathrm{p}}^2}+1.75\frac{\rho_{\mathrm{f}}\left(1-\tilde{\varepsilon}_{\mathrm{f}}\right)\left|\boldsymbol{u}_r\right|}{d_{\mathrm{p}}}, \tilde{\varepsilon}_{\mathrm{f}}<0.8\end{cases} \tag{12}$$

$$C_{\mathrm{d}}=\begin{cases}C_{\mathrm{d}}=0.44, Re_{\mathrm{d}}\geq 1000\\ \frac{24\left(1+0.15Re_{\mathrm{d}}^{0.687}\right)}{Re_{\mathrm{p}}}, Re_{\mathrm{p}}<1000\end{cases} \tag{13}$$

Where $\boldsymbol{F}_d$ is the drag force; β_d is the momentum exchange coefficient; V_p is the particle volume; $\boldsymbol{u}_r$ is the fluid-particle relative velocity; d_{p} is the particle diameter; C_{d} is the drag coefficient; is fluid viscosity; Re_p is the particle Reynolds number.

By integrating resolved, semi-resolved, and unresolved models, the hybrid approach enables differentiated coupling precision for particles of different sizes within a unified fluid grid. When computing the background flow field for medium particles, it is necessary to impose additional constraints to avoid nonphysical interactions arising from overlapping between the extended influence domain of medium particles and the fictitious fluid domain of coarse particles (Fig. 1).

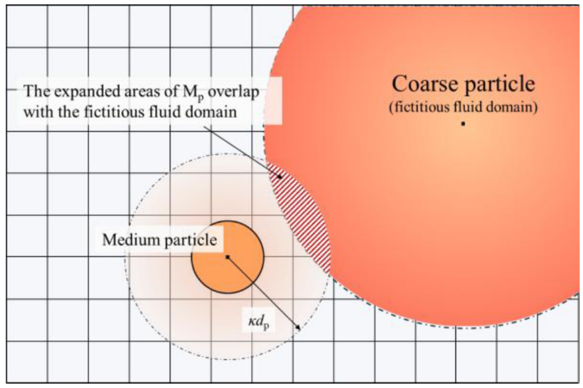


Figure 1—Overlap between the medium particle influence domain and the fictitious fluid domain of coarse particles.

A coarse particle volume fraction field is introduced to correct the porosity and fluid velocity in the background flow field for medium particles (Xie et al., 2023):

$$h = \frac{\sum_{i=1}^{n} \Delta V_{\text{cp-cover},i}}{\Delta V_{\text{cell}}} \tag{14}$$

Incorporating the coarse particle volume fraction field into the porosity calculation formula for the influence domain of medium particles yields the corrected porosity:

$$\tilde{\varepsilon}_{\text{f}}^{*} = \frac{\sum_{j}^{n} \varepsilon_{\text{cell},j}\left(1-h_j\right)\Delta V_{\text{cell},j}}{\sum_{j}^{n}\left(1-h_j\right)\Delta V_{\text{cell},j}} \tag{15}$$

Similarly, substituting the coarse particle volume fraction field into the fluid velocity formula reconstructs the background flow field velocity for medium particles:

$$\tilde{\boldsymbol{u}}_{\text{f}} = \frac{\sum_{j=1}^{n} \boldsymbol{u}\left(\boldsymbol{r}_j\right)\exp\left(-\frac{\left|\boldsymbol{r}-\boldsymbol{r}_j\right|^2}{2(\kappa d)^2}\right)\Delta V_{\text{cell},j}\left(1-h_j\right)}{\sum_{j=1}^{n}\exp\left(-\frac{\left|\boldsymbol{r}-\boldsymbol{r}_j\right|^2}{2(\kappa d)^2}\right)\Delta V_{\text{cell},j}\left(1-h_j\right)} \tag{16}$$

Where h is the coarse particle volume fraction field; n is the number of coarse particles in the grid; $\Delta V_{\text{cp-cover, j}}$, is the coverage volume of the zth coarse particle; ΔV_{cell} is the cell volume.

Particle phase

Spherical particles are used to model granular diverters, and the soft sphere model is adopted to simulate particle contacts and collisions. The inter-particle forces are calculated using the nonlinear Hertz-Mindlin model (Mindlin and Deresiewicz, 1953). Particle motion is governed by Newton's second law, and the translational and rotational equations are solved to track each particle's trajectory (Cundall and Strack, 1979):

$$m_{\text{p}}\frac{d\boldsymbol{u}_{\text{p}}}{dt} = m_{\text{p}}\boldsymbol{g} + \boldsymbol{F}_{\text{c}} + \boldsymbol{F}_{\text{f-p}} \tag{17}$$

$$\boldsymbol{I}_{\mathrm{p}}\frac{d\boldsymbol{\omega}_{\mathrm{p}}}{dt}=\boldsymbol{M}_{\mathrm{c}}+\boldsymbol{M}_{\mathrm{f\text{-}p}} \tag{18}$$

Where m_{p} is the particle mass; $\boldsymbol{F}_{\mathrm{c}}$ is the interaction force between particles or between a particle and the wall; $\boldsymbol{I}_{\mathrm{p}}$ is the moment of inertia; $\boldsymbol{\omega}_p$ is the angular velocity; $\boldsymbol{M}_c$ is the total torque acting on the particle; $\boldsymbol{M}_{\mathrm{f\text{-}p}}$ is the fluid-induced torque on coarse particles.

Flexible fibers are modeled using the bonded particle model developed and validated by Guo et al. (2013a). Adjacent spheres that constitute a fiber are connected by bonds, with the associated bond forces and torques ensuring structural integrity under external loads while providing flexibility. The motion of a fiber is determined by the dynamics of all its constituent spheres, each of which is also governed by Newton's second law:

$$m_{\mathrm{p}}\frac{d\boldsymbol{u}_{\mathrm{p}}}{dt}=m_{\mathrm{p}}\boldsymbol{g}+\boldsymbol{F}_{\mathrm{c}}+\boldsymbol{F}_{\mathrm{b}}+\boldsymbol{F}_{\mathrm{f\text{-}p}} \tag{19}$$

$$\boldsymbol{I}_{\mathrm{p}}\frac{d\boldsymbol{\omega}_{\mathrm{p}}}{dt}=\boldsymbol{M}_{\mathrm{c}}+\boldsymbol{M}_{\mathrm{b}} \tag{20}$$

Where $\boldsymbol{F}_{\mathrm{b}}$ and $\boldsymbol{M}_{\mathrm{b}}$ denote the bonding force and torque between fiber segments, respectively.

In DEM simulations, energy dissipation during particle collisions is commonly described by introducing a contact damping force. Similarly, to account for energy losses caused by elastic wave propagation during fiber deformation, the model also incorporates bond damping forces and torques. Specifically, the tangential and normal components of the bond damping force $\boldsymbol{F}_{\mathrm{d}}{}^{\mathrm{b}}$ and bond damping torque $\boldsymbol{M}_{\mathrm{d}}{}^{\mathrm{b}}$ can be expressed as follows:

$$\boldsymbol{F}_{\mathrm{d,n}}{}^{\mathrm{b}}=\beta_{\mathrm{b}}\sqrt{2m_{\mathrm{p}}K_{\mathrm{n}}^{\mathrm{b}}}\,u_{\mathrm{n}}^{\mathrm{r}} \tag{21}$$

$$\boldsymbol{F}_{\mathrm{d,t}}{}^{\mathrm{b}}=\beta_{\mathrm{b}}\sqrt{2m_{\mathrm{p}}K_{\mathrm{t}}^{\mathrm{b}}}\,u_{\mathrm{t}}^{\mathrm{r}} \tag{22}$$

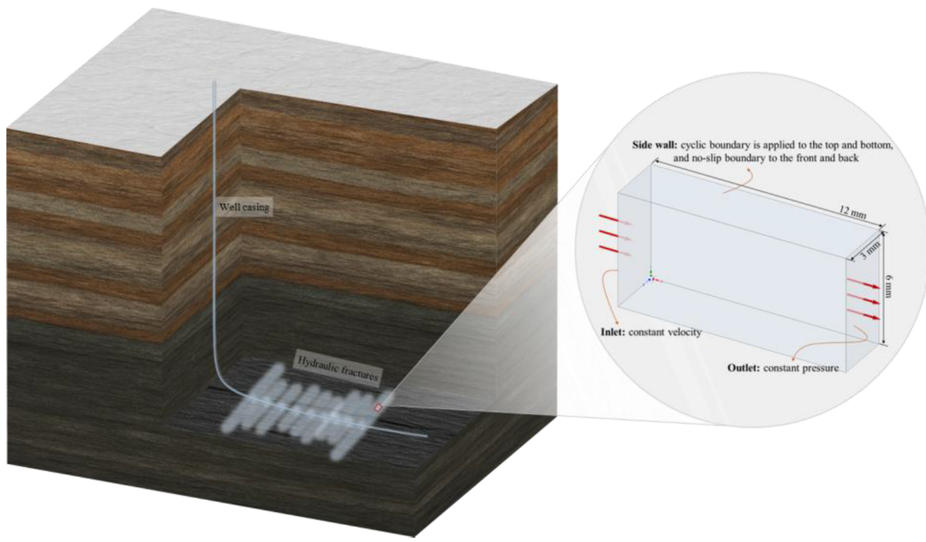
$$\boldsymbol{M}_{\mathrm{d,n}}{}^{\mathrm{b}}=\beta_{\mathrm{b}}\sqrt{2I_{\mathrm{p}}K_{\mathrm{tor}}^{\mathrm{b}}}\,\omega_{\mathrm{n}}^{\mathrm{r}} \tag{23}$$

$$\boldsymbol{M}_{\mathrm{d,t}}{}^{\mathrm{b}}=\beta_{\mathrm{b}}\sqrt{2I_{\mathrm{p}}K_{\mathrm{ben}}^{\mathrm{b}}}\,\omega_{\mathrm{t}}^{\mathrm{r}} \tag{24}$$

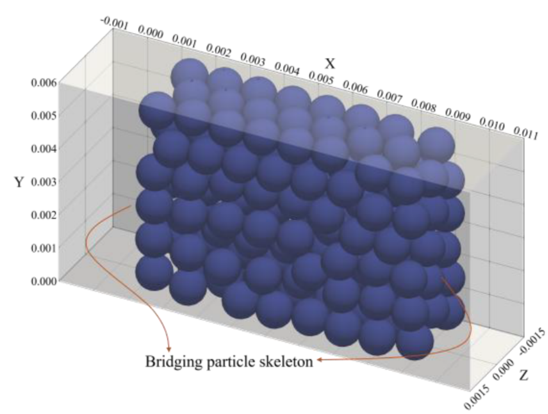
Where β_{b}, is the bond damping coefficient, which controls the energy dissipation rate; *U* and *(O′* are the relative translational and rotational velocities between bonded particles; $K_{\mathrm{n}}{}^{\mathrm{b}}$, $K_{\mathrm{t}}{}^{\mathrm{b}}$, $K_{\mathrm{tor}}{}^{\mathrm{b}}$, and $K_{\mathrm{ben}}{}^{\mathrm{b}}$ are the normal, shear, torsional, and bending stiffnesses of the bond, respectively.

Numerical model

The computational domain of the model is a rectangular region measuring 12 × 3 × 6 mm, representing a small section of a hydraulic fracture, as illustrated in Fig. 2 (a). The fluid mesh element size is 0.1 mm, with a total of 216,000 grid cells throughout the domain. The grid size to bridging particle diameter ratio is 0.1; an analytical model is used to resolve the pore-scale seepage flow field within the bridging skeleton. The front and rear surfaces of the model represent fracture walls, with no-slip boundary conditions applied. To minimize the influence of the finite domain on fluid flow, periodic boundary conditions are applied at the upper and lower surfaces. The fluid, carrying particles, enters from the left at a constant velocity, while a fixed pressure outlet is set at the right boundary. The standard k-*e* model is used to capture fluid turbulence. During temporary plugging operations in the Weiyuan block shale gas wells in the Sichuan Basin, the single fracture injection rate is typically 0.3~0.75 m^3/min, corresponding to an average in-fracture fluid velocity of 0.02~0.06 m/s. Since this study aims to investigate the formation mechanism of dense temporary plugging layers at the fracture tip, the fluid velocity in the simulations is set slightly lower than the field average, at 0.01 m/s. The parameters for diverter particles and fibers are selected based on typical values reported in the literature (Ghommem et al., 2020; Henriques and Dahi Taleghani, 2024). Detailed parameter values are provided in Table 1.



(a)



(b)

Figure 2—Numerical model: (a) geometry of the computational domain; (b) randomly generated bridging particle skeleton.

Table 1—Parameters of the numerical model.

Model	Parameter	Value
Computational Domain	Length, m	0.012
	Width, m	0.003
	Height, m	0.006
Fluid	Density, kg/m^3	1000
	Viscosity, Pa's	0.005
	Velocity, m/s	0.01
Particle	Density, kg/m^3	1300
	Young's modulus, GPa	2.6
	Poisson's ratio	0.3
	Coefficient of restitution	0.5
	Coefficient of friction	0.4

Model	Parameter	Value
Fiber	Density, kg/m³	1300
	Young's modulus, GPa	0.1
	Poisson's ratio	0.2
	Coefficient of restitution	0.3
	Coefficient of friction	0.61

Before the simulation begins, 150 bridging particles with a diameter of 1 mm—a typical size for temporary plugging agents in hydraulic fractures—are randomly generated within the domain. The upstream and downstream filter grids are then moved toward the center to compact the bridging particle pack. These filter grids confine the movement of bridging particles while allowing fine particles and fibers to pass through. The resulting bridging skeleton, shown in Fig. 2(b), exhibits a porosity of approximately 45%. In the simulation, fine particles or fibers are generated at $x = 0$ and assigned an initial velocity of 0.01 m/s.

The hybrid CFD-DEM approach is implemented using the open-source software OpenFOAM, LIGGGHTS, and CFDEMCoupling, with CFDEMCoupling dynamically selecting the coupling algorithm and managing data exchange between fluid and solid phases based on the grid-to-particle size ratio. The specific numerical implementation workflow is illustrated in Fig. 3.

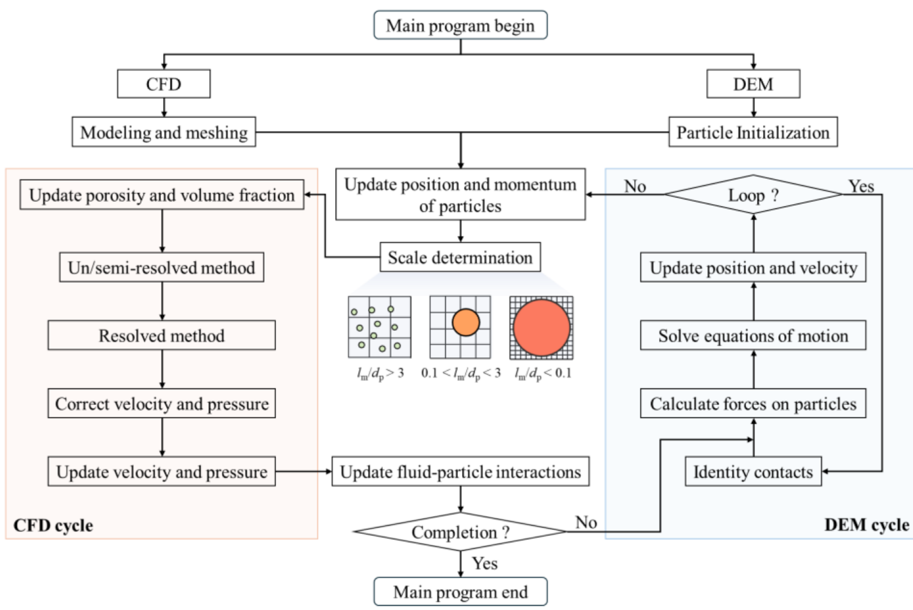


Figure 3—Solution procedure of the hybrid CFD-DEM.

To ensure computational accuracy and efficiency during the hybrid CFD-DEM solution process, only the Courant number and Rayleigh time constraints are considered when only particles are present (Li et al., 2005; Mondal et al., 2016), with the CFD time step set to $1x10^{-5}$ s, the DEM time step set to $5x10^{-7}$ s, and the coupling interval set to 20. When fibers are present, the DEM time step must also be smaller than the time required for axial waves to propagate through a bond (Guo et al., 2013b). In this case, the CFD time step is set to $1x10^{-6}$ s, the DEM time step to $5x10^{-8}$ s, and the coupling interval is maintained at 20.

Results and discussion

Effect of particle size

The migration and clogging behaviors of three types of fine particles with different diameters, as well as a mixed particle system composed of these three types at a 1:1:1 ratio (all at a concentration of 60 kg/m^3), were investigated within the bridging skeleton. To quantify differences in transport capacity, the average horizontal migration velocity was measured, as shown in Fig. 4(a). Based on the inlet fluid velocity and the porosity of the bridging skeleton, the initial average fluid velocity in the pores is calculated to be approximately 0.022 m/s. At the initial stage of injection, the average migration velocity of fine particles significantly exceeded both the initial injection velocity and the average pore fluid velocity, indicating the presence of preferential flow paths within the skeleton that promote inertial particle migration. At this stage, the differences in average migration velocity among fine particles of different sizes are negligible. As injection progresses, the enhanced mobility associated with smaller particle sizes gradually becomes evident. The final average migration velocity of the 0.10 mm particles is higher than that of the 0.13 mm and 0.15 mm particles, suggesting that smaller particles are more mobile and less prone to clogging at pore throats. In contrast, larger fine particles tend to form clogs within the skeleton (Fig. 4(b)), impede the transport of subsequent particles, and cause a rapid decline in average migration velocity. The average migration velocity and clogging ratio of the mixed particle system are close to those of the 0.13 mm particles, indicating that the presence of small particles in the mixture alleviates clogging at pore throats and enhances overall mobility, while large particles promote the formation of local clogs.

In terms of clogging distribution, gravity causes the vast majority of clogging to occur in the lower part of the skeleton, especially near the outlet side of the model, with only sporadic clogging observed in the upper right corner of the skeleton (Fig. 4(c)). Across the four cases, the morphology of the local clogging zones formed by fine particles is similar, though the degree of clogging differs. This variation is related to the pore structure within the skeleton, where dispersed narrow throats are more likely to capture fine particles. Larger fine particles tend to form size-dominated pore clogs, resulting in pronounced pore contraction and throat narrowing, which enhance the interception of subsequent particles. For the mixed particle system, fine particles are able to migrate deeper into the skeleton and, compared to the single-size cases, their distribution throughout the pore space is more uniform. The local average normal contact force in clogging zones is normalized by the global average, as illustrated in Fig. 4(d), with each bar representing the ratio of the interval averaged normal contact force to the global average. Analysis of the normal contact forces in the clogging zones reveals that the anisotropy of the normalized local average normal contact force increases with decreasing particle size, whereas this anisotropy is significantly reduced in the mixed particle system. This is attributed to the cooperative participation of particles of different sizes in clogging, where small particles fill the gaps between large particles, resulting in a more uniform distribution of contact forces and thus mitigating anisotropy.

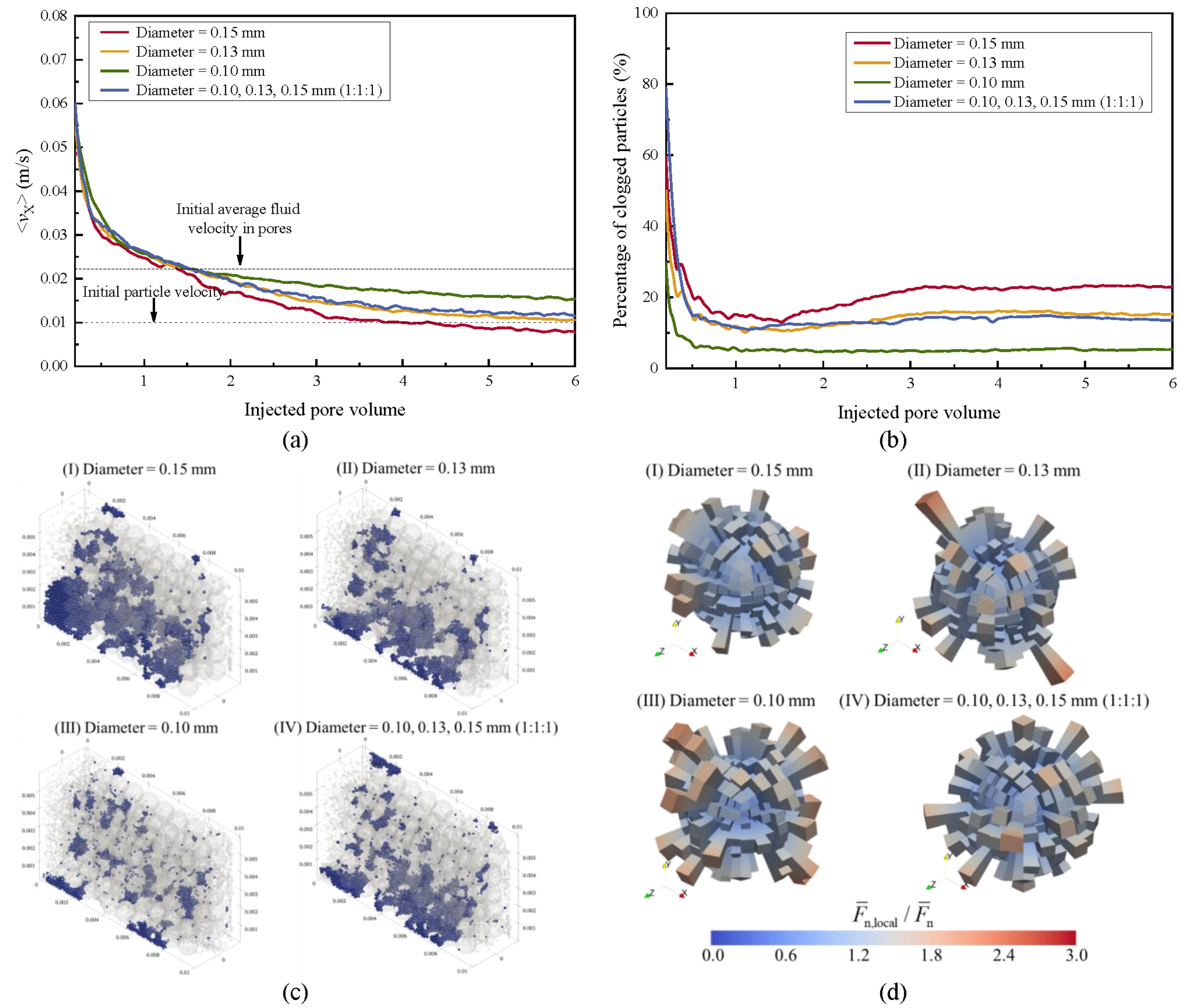


Figure 4—Migration and clogging characteristics of fine particles and fibers in the bridging skeleton: (a) average migration velocity; (b) evolution of clogging ratio; (c) spatial distribution of clogged particles and fibers at the end of injection; (d) standardized local average normal contact force distribution.

Effect of concentration

Taking fine particles with a diameter of 0.13 mm as an example, the migration and clogging behavior swithin the bridging skeleton were analyzed under two different concentrations (30 kg/m^3 and 60 kg/m^3). As shown in Fig. 5(a), during the initial injection stage, the average migration velocities for both low and high concentration cases were similar and both exceeded the initial injection velocity, exhibiting pronounced inertial migration characteristics. With increasing injected pore volume, particularly after approximately two pore volumes, the average migration velocity of the high-concentration particles declined significantly, falling below that of the low-concentration case. This was mainly attributed to intensified competition at pore throats under high concentration, which increases the likelihood of clogging and thus suppresses further particle migration. The evolution of the clogging ratio with injected volume is illustrated in Fig. 5(b). In the early stage, the clogging ratio of high-concentration particles is lower than that of low-concentration particles, possibly because dominant flow paths have not yet been blocked, resulting in substantial particle loss. However, as the clogging structure develops and these dominant channels are gradually plugged, new clogging preferentially forms in the remaining channels under high- concentration conditions. As a result, the clogging ratio for high-concentration particles eventually surpasses that of low-concentration particles, achieving higher plugging efficiency in the later stages.

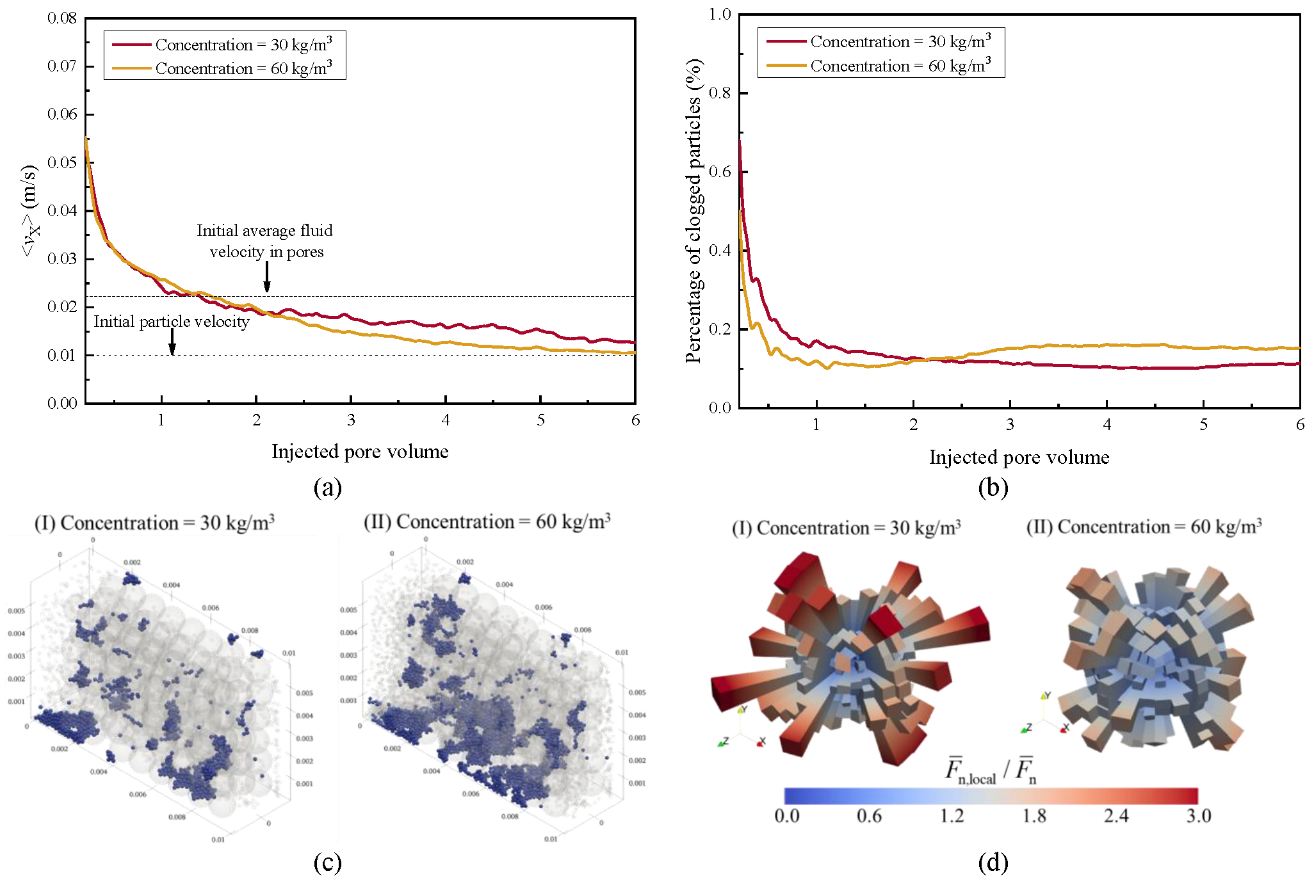


Figure 5—Migration and clogging characteristics of fine particles and fibers in the bridging skeleton: (a) average migration velocity; (b) evolution of clogging ratio; (c) spatial distribution of clogged particles and fibers at the end of injection; (d) standardized local average normal contact force distribution.

The spatial distribution of clogged particles at the end of injection (Fig. 5(c)) indicates that, under high concentration, the lower region of the skeleton exhibits more intensive clogging, while distinct local clogging structures also gradually form in the upper region. This suggests that higher concentration promotes more comprehensive and uniform clogging across the skeleton. Additionally, the spatial distributions of the normalized normal contact force under both concentration conditions are shown in Fig. 5(d). At low concentration, due to fewer and more dispersed local clogging regions, the normalized normal contact force exhibits pronounced anisotropy. As particle concentration increases, the number of particles within clogged regions rises and particle-particle interactions intensify, leading to a reorganization of the flow field and a progressive weakening of directionality, as evidenced by reduced anisotropy of the normal contact force. Overall, increasing particle concentration not only facilitates the formation of clogging structures but also enhances the mechanical stability of the clogged regions.

Effect of diverters' shape

The migration and clogging behaviors of fine particles and fibers within the bridging skeleton were comparatively analyzed under identical concentration conditions (30 kg/m^3). The fine particles had a diameter of 0.13 mm, while the fibers were constructed from 15 spherical particles with a diameter of 60 gm each, using a bonded particle model. As shown in Fig. 6(a), despite the small size of the constituent particles, the overall average migration velocity of fibers is consistently and markedly lower than the initial injection velocity, significantly lower than that of fine particles, and rapidly decays to nearly zero. This result indicates that fibers, due to their high aspect ratio, are strongly restricted during migration through the pore channels of the skeleton, making long-distance transport difficult and leading to a tendency for retention at the skeleton entrance. The evolution of the clogging ratio with injected pore volume is illustrated in Fig. 6 (b). In the middle and late stages of injection, the clogging ratio of fibers exceeds 80%, whereas the clogging ratio for fine particles remains at approximately 15% throughout. The spatial distribution of the clogged regions (Fig. 6 (c)) further highlights this contrast: fibers mainly form extensive clogging at the skeleton entrance, predominantly as U-shaped structures that attach and entangle on the surfaces of bridging particles, which substantially impedes the deep migration of subsequent fibers. In contrast, the local clogs formed by fine particles are more dispersed within the skeleton, resulting in a lower overall clogging ratio.

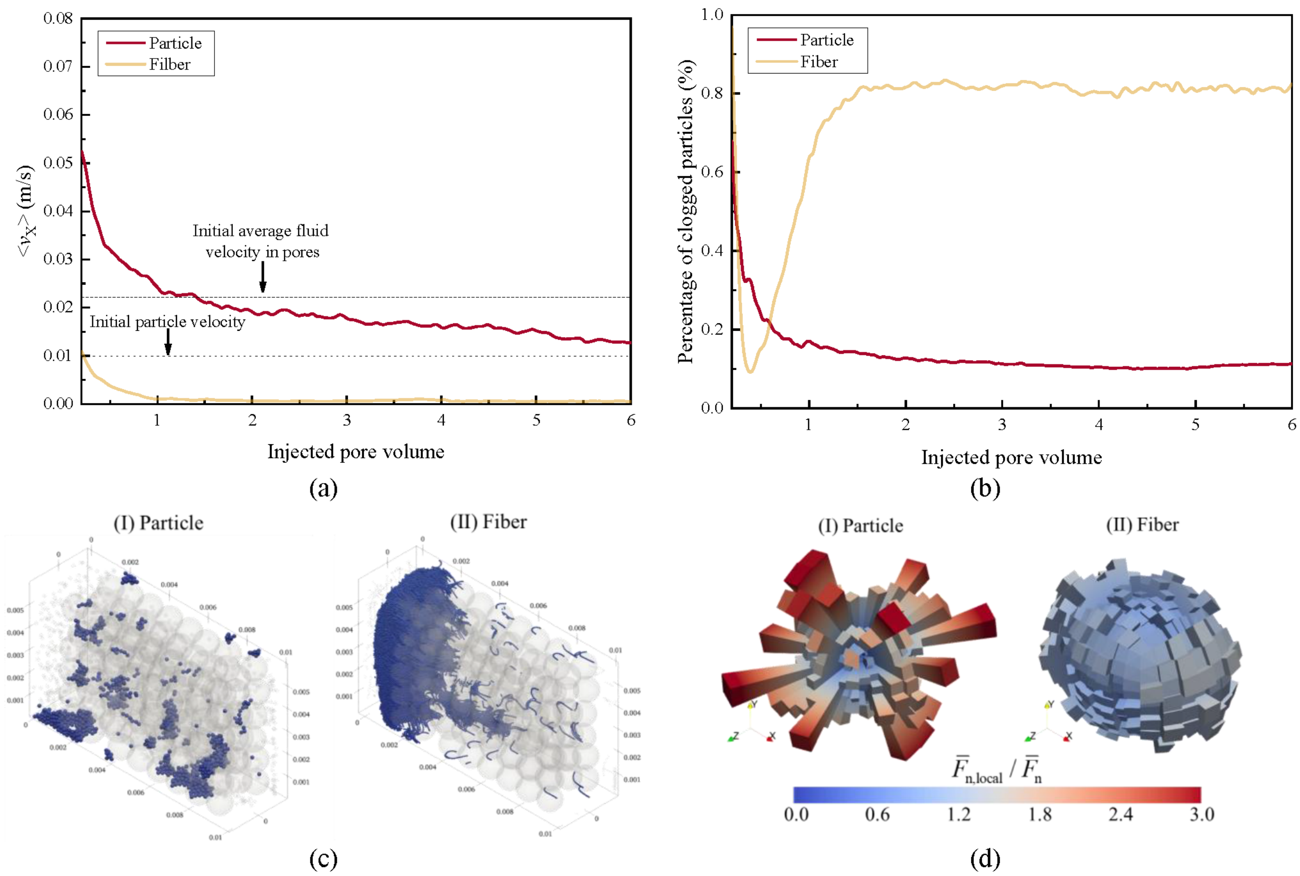


Figure 6—Migration and clogging characteristics of fine particles and fibers in the bridging skeleton: (a) average migration velocity; (b) evolution of clogging ratio; (c) spatial distribution of clogged particles and fibers at the end of injection; (d) standardized local average normal contact force distribution.

Fig. 6 (d) shows that the distribution of normalized normal contact forces in the fiber case is nearly isotropic. This phenomenon arises because, after entering the skeleton region, fibers are no longer merely aligned by fluid flow but become embedded and constrained within the evolving clogging structure. Fibers covering the surfaces of bridging particles or the clogging layer are restricted by multiple contact points with the surroundings. As the clogging structure becomes denser, fibers further bend and wrap around granular or pore- scale features. These spatial constraints and mechanical interlocking diminish directional dominance, resulting in a more uniform distribution of normal contact forces. This indicates that fiber-dominated sealing layers not only efficiently block the skeleton entrance but also disperse fluid impacts more uniformly, thereby enhancing the structural stability and pressure-bearing capacity of the sealing layer.

Conclusions

This study systematically investigates the migration and clogging behaviors of diverters with various particle sizes, concentrations, and shapes (particles and fibers) within a bridging skeleton, employing a hybrid CFD-DEM approach. The research clarifies the multiscale mechanisms involved in dense sealing layer formation. The main conclusions are as follows:

1. Fine particles initially migrate rapidly along preferential flow paths within the skeleton. Smaller particles demonstrate greater mobility and are less likely to clog pore throats, while larger particles accumulate and form size-dominated local clogs that hinder subsequent transport. In mixed particle systems, small particles help relieve clogging and promote more uniform distribution, whereas large particles contribute to denser local sealing.
2. Higher particle concentrations increase competition at pore throats, leading to faster formation of dense and stable clogging structures, particularly in the lower part of the skeleton. This enhances both plugging efficiency and mechanical stability.
3. Fibers, despite being composed of small-diameter spheres, exhibit limited transport due to their high aspect ratio. They preferentially accumulate at the skeleton entrance and form extensive U-shaped clogging zones, which improve sealing performance at the inlet.
4. Analysis of normal contact force distribution shows pronounced anisotropy in local force chains for low concentration, singlesize particle systems. In contrast, mixed particle systems, higher concentrations, and fiber addition all reduce anisotropy, thereby improving the structural stability and pressure-bearing capacity of the clogged regions.

This work elucidates the synergistic mechanisms of diverters migration and clogging in bridging regions, providing theoretical guidance for optimizing temporary plugging strategies and material design in hydraulic fracturing. The developed hybrid CFD-DEM approach also supports numerical simulation of complex multiscale particle-fluid systems.

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